Fast Fourier Transform "FFt"

For N = 4 samples/cycle.

* Calculate the FFt of data sequance
  $x(n) = \{3, 1, 1, 2\}$
* check your Results using iFFt
* Plot the Amplitude spectrum
* Plot The Phase spectrum
* Plot The Energy spectrum
* check your Results using Parseval's theorem.

→ Compare between the DFt, FFt According to Complex Addition, Complex multiplication

→ Computional Saving

→ Prove that the DFt, X(K) of a sequance x(n) is Periodic

mid → Find The Relation between Ft, DFt.

$$FT = T_s * DFt$$

$x(n) = \{ \overset{x(0)}{3}, \overset{x(1)}{1}, \overset{x(2)}{1}, \overset{x(3)}{2} \}$

using FFt

- re arrange the samples.

$x(0)$	Binary → $x(00)$	mirror → $x(00)$	Decimal → $x(0)$
$x(1)$	→ $x(01)$	→ $x(10)$	→ $x(2)$
$x(2)$	→ $x(10)$	→ $x(01)$	→ $x(1)$
$x(3)$	→ $x(11)$	→ $x(11)$	→ $x(3)$

- ToTal Number of stage $m = \log_2 N$

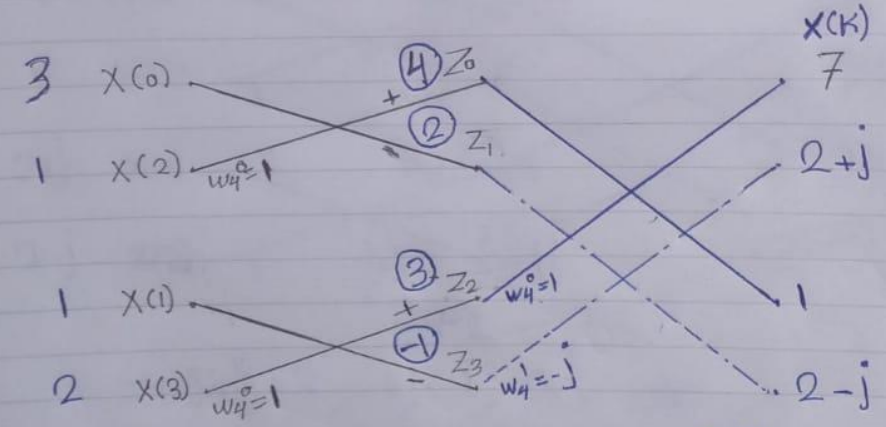
$= \log_2 2^2 = 2\log_2 2 = 2$

- Number of Butter Fly in each stage

$\frac{N}{2} = 2$

- The Twiddle Factor

stage 1	stage 2
$W_4^0 = 1$	$W_4^0 = 1$
$W_4^0 = 1$	$W_4^1 = -j$



$x(K) = \{ 7, 2+j, 1, 2-j \}$

For stage 1 eQuation For FFt

$Z_0 = X(0) + W_4^0 X(2) = 3 + 1 \cdot 1 = 3+1 = 4$

$Z_1 = X(0) - W_4^0 X(2) = 3 - 1 \cdot 1 = 3-1 = 2$

$Z_2 = X(1) + W_4^0 X(3) = 1 + 1 \cdot 2 = 1+1 = 3$

$Z_3 = X(1) - W_4^0 X(3) = 1 - 1 \cdot 2 = 1-2 = -1$

For stage 2

X(K)

$X(0) = Z_0 + W_4^0 Z_2 = 4 + 1 \cdot 3 = 7$

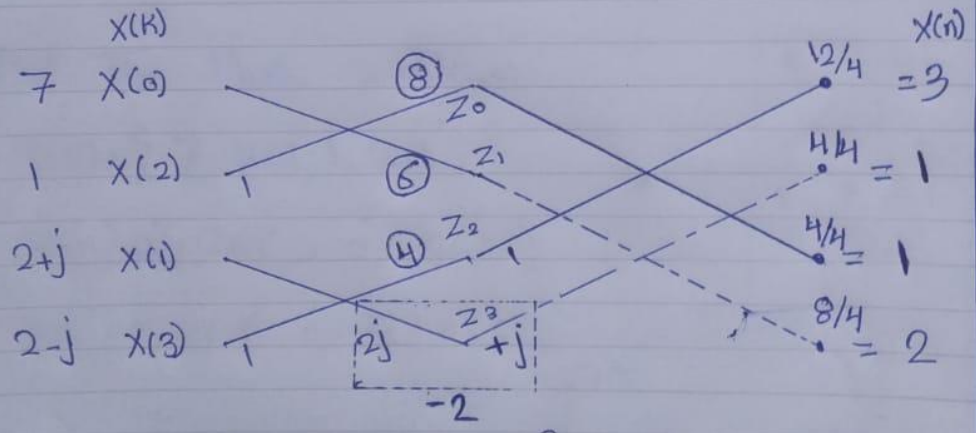
$X(1) = Z_1 + W_4^1 Z_3 = 2 + j * 1 = 2+j$

$X(2) = Z_0 - W_4^0 Z_2 = 4 - 3 = 1$

$X(3) = Z_1 - W_4^1 Z_3 = 2 - j \cdot 1 = 2-j$

the inverse of Fast Fourier Transform "iFFt"

→ $X(K) = \{7, 2+j, 1, 2-j\} \Rightarrow$ Get $X(n) = ??$



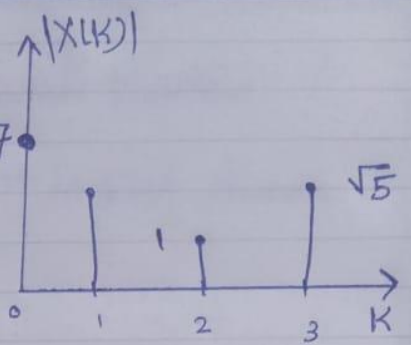
$X(n) = \{3, 1, 1, 2\}$ #

[4]

*Plot The Amplitude spectrum

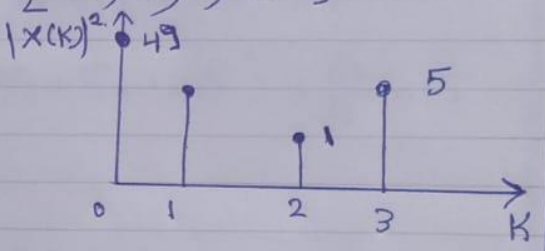
$X(K) = \{7, 2+j, 1, 2-j\}$

Amplitude $= |X(K)| = \{7, \sqrt{5}, 1, \sqrt{5}\}$



* Plot The Energy spectrum

Energy spectrum $= |X(K)|^2 = \{49, 5, 1, 5\}$



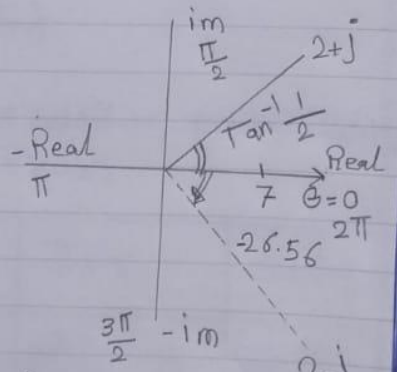
* Plot The Phase spectrum

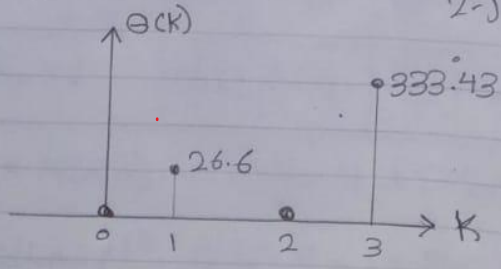
$X(K) = \{7, 2+j, 1, 2-j\}$

$\theta(K) = \{0^\circ, 26.6^\circ, 0^\circ, -26.56^\circ\}$

or

$\theta(K) = \{0, 26.6, 0^\circ, 333.43\}$





* check your Results using Parseval's theorem

Energy in Time domain = Energy in Freq domain

$$\sum_{n=0}^{N-1} x(n)^2 = \frac{1}{N} \sum_{K=0}^{N-1} |X(K)|^2$$

$$L.H.S = \sum_{n=0}^{3} x(n)^2 = x(0)^2 + x(1)^2 + x(2)^2 + x(3)^2 =$$
$$3^2 + 1^2 + 1^2 + 2^2 = 15$$

$$R.H.S = \frac{1}{N} \sum_{K=0}^{N-1} |X(K)|^2 = \frac{1}{4}\left[|X(0)|^2 + |X(1)|^2 + |X(2)|^2 + |X(3)|^2\right]$$
$$= \frac{1}{4}[49 + 5 + 1 + 5] = \frac{60}{4} = 15$$

∴ L.H.S = R.H.S

∴ The Parseval's theorem is Verified

→ Compare between DFt, FFt (Complex Addition, multiplication) or Computional Saving Process

	DFt	FFt	Saving
Complex Addition	$N(N-1) = 4 \cdot 3 = 12$	$N \log_2 N = 8$	4 Process
Complex multiplication	$N^2 = 16$	$\frac{N}{2} \log_2 N = 4$	12 Process

Saving in Complex Add = $N(N-1) - N \log_2 N = 4$ Process

in // multiplication = $N^2 - \frac{N}{2} \log_2 N = 12$ Process

→ Prove that the DFt, X(K) of sequance X(n) is Periodic (imp)

the DFt

$$X(K) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi Kn}{N}}, \quad K = 0 \cdots N-1$$

For the Periodicity of x(n)

$$x(n+N) = x(n) \qquad n \longrightarrow n+N$$

For the Periodicity of X(K)

$$X(K+N) \stackrel{?}{=} X(K) \qquad K \longrightarrow K+N$$

Let replace

$$K \longrightarrow K+N$$

$$X(K+N) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi n}{N}(K+N)}$$

$$= \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi nK}{N} - j\frac{2\pi n.N}{N}} \qquad e^{x+y} = e^{x}.e^{y}$$

$$= \sum x(n)\, e^{-j\frac{2\pi Kn}{N}} \cdot e^{-j2\pi n}$$

$$= \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi Kn}{N}} (\cos 2\pi n - j\sin 2\pi n)$$

$$(\cos 2\pi n \to 1, \quad \sin 2\pi n \to 0)$$

$$= \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi Kn}{N}} \cdot 1 = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi Kn}{N}} = X(K)$$

$\because X(K+N) = X(K)$

$\therefore$ X(K) is Periodic signal with Fundamential Period N